

Dynamical systems and ergodic theory

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¹Based on the lessons by professor Marco Lenci

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Main definitions

Fundamental definition

Definition 0.1 A dynamical system (DS) over the set M (usually called phase space or reference space) is a family $\phi^t_{t \in \mathbb{G}}$ of transformations $\phi^t : M \rightarrow M$ with the (semi)group property:

$$\begin{aligned}\phi^0 &= id_M \\ \phi^{t+s} &= \phi^s \circ \phi^t\end{aligned}$$

Note: $\mathbb{G}$ must be a commutative semigroup with neutral element with respect to the "+" operation. In practice, for the contents of this course, $\mathbb{G} = \mathbb{R}, \mathbb{R}_0^+, \mathbb{Z}, \mathbb{N}$

Observation 0.1 Differential equations not depending on time derivatives satisfy the definition.

Usually the formalism is that, say x is the state now (meaning at $t = 0$), then the state at time t is given by $\phi^t(x)$.

Definition 0.2 If $\mathbb{G} = \mathbb{R}, \mathbb{R}_0^+$, then ϕ^t is called the flow of the DS. if, instead, $\mathbb{G} = \mathbb{Z}, \mathbb{N}$ usually you rename it as $T = \phi^1$, $n = t$ where T is called a map.

Observation 0.2 This is due to the fact that by just knowing T you can have enough information to know the future of the system, since

$$\phi^t = (\phi^1)^n = \underbrace{\phi^1 \circ \phi^1 \dots \circ \phi^1}_{n \text{ times}} = T^n$$

Note that from now on if n is an integer exponent it means to use the transformation n times. Also, if T is invertible (bijective) we denote, for $n \in \mathbb{Z}^+$:

$$T^{-n} = \underbrace{T^{-1} \circ T^{-1} \dots \circ T^{-1}}_{n \text{ times}}$$

Observe that $\forall n, m \in \mathbb{Z}$:

$$T^{n+m} = T^n \circ T^m$$

Notice that by having T be invertible, it means that by knowing it you can also go back in time, and as such you know everything about the system; we will see both bijective and not in this course. (here he drew a figure).

Chapter 1

Examples of simple dynamical systems

1.1 Important example: autonomous ODEs

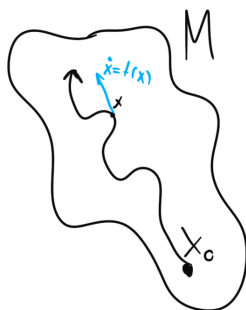


Figure 1.1: Representation of the quantities in an autonomous ODE.

The most obvious example of a dynamical system is given by an ODE, which takes the form of:

$$\dot{x} = f(x)$$

where $M \subseteq \mathbb{R}^d$, $f : M \rightarrow \mathbb{R}^d$.

We are looking at a curve of which we know the first time derivative, and we know said derivative to be a function of position, but not of time, as shown in Figure 1.1.

The aforementioned problem has, potentially, infinite solutions. What interests us are what are called Cauchy problems or, more commonly,¹ IVP, Initial

¹This is true for the whole world, apart from Italy and France, which only call those Cauchy problems. The latter has obvious reasons to do so. Italy, on the other hand, hasn't got so many.

Value Problems.

Definition 1.1 (Cauchy problem) A Cauchy problem is given by the following linear system:

$$\begin{cases} \dot{x} = f(x) \\ x(0) = x_0 \end{cases} \quad (1.1)$$

Cauchy's theorem tells us that under some conditions, which we assume, the solution exists, and it's unique. (those conditions being that $f(x)$ is a Lipschitz function).

We shall start with the classic ODE example: the harmonic oscillator.

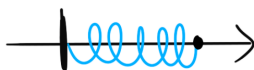


Figure 1.2: Representation of a harmonic oscillator.

Example 1.1 (Harmonic oscillator) The one dimensional harmonic oscillator is a physical system in which a mass is attracted to the origin by a force that depends on the distance to the origin itself, as shown in Figure 1.2. This is described by:

$$\ddot{x} = -w^2x \quad (1.2)$$

At first glance, it may look like it does not fit the definition of a Cauchy problem, since it is a second derivative instead of a first one. Turns out, all high order ODEs may be reduced to a single order one.

The trick is to rename the first derivative. in this case:

$$\begin{cases} \dot{x} = v \\ \dot{v} = -w^2x \end{cases}$$

Or, in matrix form:

$$\begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} v \\ -w^2x \end{bmatrix} = f\left(\begin{bmatrix} x \\ v \end{bmatrix}\right)$$

We did increase the dimension of the phase space, but that does not get in the way of calling this a Cauchy problem.

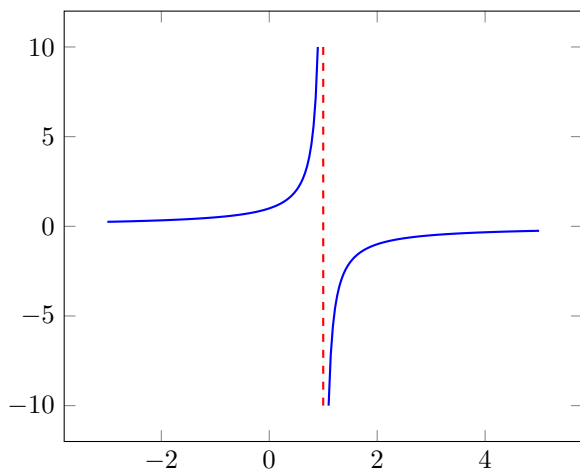
Other than uniqueness and existence, we will also be implicitly assuming all solutions to be global, i.e. defined $\forall t \in \mathbb{R}$, which is a much more restrictive condition. While finding a problem for which uniqueness and existence are not true is quite hard, the same is not true for a Cauchy problem without a global solution. Case in point:

Example 1.2 (of a non global solution)

$$\begin{cases} \dot{x} = x^2 \\ x(0) = 1 \end{cases}$$

where $M = \mathbb{R}$. Solving it by abusing math a bit:

$$\frac{dx}{dt} = x^2 \Leftrightarrow \frac{dx}{x^2} = dt \Leftrightarrow \int_1^x \frac{d\xi}{\xi^2} = \int_0^t d\tau \Leftrightarrow \left[-\frac{1}{\xi} \right]_1^x = t \Leftrightarrow 1 - \frac{1}{x} = t \Leftrightarrow x = \frac{1}{1-t}$$



As you can see, it blows up asymptotically at 1, so it is not a global solution.

We shall now see the concept of flow.

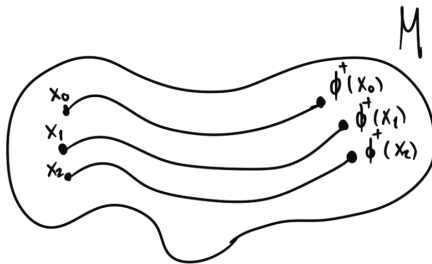


Figure 1.3: Representation of flow.

Definition 1.2 If $x(t; x_0)$ is the solution of Equation 1.1, the flow of such system is:

$$\phi^t(x_0) = x(t; x_0) \quad (1.3)$$

Proposition 1.1 $\phi_{t \in \mathbb{R}}^t$, as defined in Equation 1.3 satisfies the group property.

Proof. The group properties are two:

$$\phi^0 = id_M \quad (1.4)$$

$$\phi^{t+s} = \phi^s \circ \phi^t \quad (1.5)$$

Equation 1.4 is trivial, because by definition $\phi^0(x_0) = x_0 \forall x_0 \in M$.

The second property is a bit harder. First of all, by definition:

$$\phi^{t+s} = \phi^t \circ \phi^s \Leftrightarrow \phi^t(\phi^s(x_0)) = \phi^{t+s}(x_0) \forall x_0 \quad (1.6)$$

We define $y_1(t) = \phi^t(\phi^s(x_0))$ and $y_2(t) = \phi^{t+s}(x_0)$. We'll prove that $y_1(t)$ and $y_2(t)$ are both solutions of the following IVP:

$$\begin{cases} \dot{y}(t) = f(y(t)) \\ y(0) = \phi^s(x_0) \forall s \in \mathbb{R}, x_0 \in M \end{cases} \quad (1.7)$$

Because of Cauchy's theorem of uniqueness, if both solve the IVP, they must be equal, thus the proposition will be proved because of Equation 1.6.

The initial condition is true by definition:

$$\begin{aligned} y_1(0) &= \phi^0(\phi^s(x_0)) = \phi^s(x_0) \\ &\Leftrightarrow y_1(0) = y_2(0) = \phi^s(x_0) \forall s \in \mathbb{R} \\ y_2(0) &= \phi^{0+s}(x_0) = \phi^s(x_0) \end{aligned}$$

As for the other condition, for y_1 it's true by definition, first by the definition of y_1 itself, then by the definition of $\phi^t(x_0)$:

$$\begin{aligned} \frac{dy_1}{dt}(t) &= \frac{d\phi^t}{dt}(\phi^s(x_0)) = \frac{dx(t; \phi^s(x_0))}{dt} = \\ f(x(t; \phi^s(x_0))) &= f(\phi^t(\phi^s(x_0))) = f(y_1(t)) \end{aligned}$$

where x is the solution of the IVP over which the flow is defined, while $f()$ for which $\dot{x} = f(x)$ is true.

In the same way, by employing the definitions of y_2 and flow, aided by the chain rule, we may find that the temporal derivative of y_2 is equal to $f(y_2)$, where $f()$ is the same as the one for which that is true for y_1 :

$$\begin{aligned} \frac{dy_2}{dt}(t) &= \frac{d\phi^{t+s}}{dt}(x_0) = \frac{dx(t+s; x_0)}{d(t+s)} \frac{d(t+s)}{dt} = \\ f(x(t+s; x_0)) &= f(\phi^{t+s}(x_0)) = f(y_2(t)) \end{aligned}$$

So, y_1 and y_2 both solve the aforementioned IVP for every t , thus we proved the proposition. □

Let's now look a bit into the concept of flow. The flow behaves in such a way (this is why it's called a flow!) in which it never intercepts. It looks, if drawn in a stylized way, as a series of lines that never intercept, because of the uniqueness of the solution of the Cauchy problem, as shown in Figure 1.3.

Proposition 1.2 *We define two flows $\phi^{t_1}(x_1)$ and $\phi^{t_2}(x_2)$. If $t_1 \geq t_2$ and $x_1 \neq x_2$, they shall never intercept.*

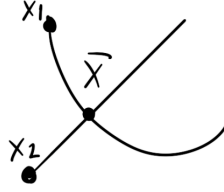


Figure 1.4: Representation of the hypothesis used to prove Proposition 1.2.

Proof. RAA² Assume that we have two trajectories that are completely distinct apart from one singular interception. Let's call the singular interception $\bar{x}$. The following is true by definition:

$$\bar{x} = \phi^{t_1}(x_1) = \phi^{t_2}(x_2)$$

But then, by definition of flow:

$$\phi^{-t_2}(\bar{x}) = \phi^{-t_2}(\phi^{t_2}(x_2)) = x_2$$

but at the same time:

$$\phi^{-t_2}(\bar{x}) = \phi^{-t_2}(\phi^{t_1}(x_1)) = \phi^{t_1-t_2}(x_1)$$

The hypothesis here that the interception is just one. Starting from said hypothesis, we found another interception, namely x_2 . This would mean, because of the uniqueness of the solution of the Cauchy problem, that the trajectories are exactly the same: but that is impossible, because we assumed that they were distinct apart from one singular point. $\square$

Now: keep in mind that a global solution is not hard to imagine.

Example 1.3 (of global solution ascertained by visual inspection) Suppose M is a two-dimensional domain and suppose you know that at the boundary the vector field f always points inside. And suppose you know that this vector field is smooth and defined inside all the domain. Then you are assured that the solution is global. This because the vector field cannot "escape" the domain, and it covers all of it.

Usually you can find a global solution by iteratively move inside small intervals.

Now, for a little result that is more of a curiosity than anything else.

Proposition 1.3 *If ϕ^t , a continuous DS on $M \subseteq \mathbb{R}^d$, is smooth enough³, then it is the flow of this ODE:*

$$\dot{x} = \left. \frac{\partial \phi^t}{\partial t} \right|_{t=0}$$

²Reductio ad absurdum. I will use this to signal that I am proving by contradiction.

³Meaning C^1 Lipschitz, in this case.

Proof. As it is a continuous DS, it has to satisfy the semi-group property.

Keeping this in mind, to verify the proposition we need to define the function

$$x(t, x_0) : t \rightarrow \phi^t(x_0)$$

for all $x_0 \in M$, then verify that it satisfies the ODE, given the initial condition:

$$x(0, x_0) = x_0$$

The initial condition is true by definition, because $x(0, x_0) = \phi^0(x_0) = x_0$.

Then, we must verify that it does indeed solve the ODE. By definition:

$$\dot{x}(t, x_0) = \frac{\partial \phi^t(x_0)}{\partial t}$$

then, by employing the chain rule⁴:

$$\begin{aligned} \frac{\partial \phi^t(x_0)}{\partial t} &= \frac{\partial \phi^{t+s}(x_0)}{\partial s} \Big|_{s=0} = \frac{\partial}{\partial s} \phi^s(\phi^t(x_0)) \Big|_{s=0} = \\ &= \frac{\partial \phi^s(\phi^t(x_0))}{\partial s} \Big|_{s=0} = \frac{\partial \phi^s(x(t, x_0))}{\partial s} \Big|_{s=0} \end{aligned}$$

Then, straightforwardly:

$$\dot{x}(t, x_0) = \frac{\partial \phi^s(x(t, x_0))}{\partial s} \Big|_{s=0} = \frac{\partial \phi^t}{\partial t}(x(t, x_0)) \Big|_{t=0}$$

As ugly as that may be, it is true, since s is just a name: we can change s to t without second thoughts because it is a partial derivative, which fixes $x(t, x_0)$ in place, thus avoiding uncomfortable situations with the temporal dependency of the position. $\square$

1.2 Non-autonomous ODEs

Definition 1.3 A non-autonomous ODE is a differential equation of the form:

$$\dot{x} = f(x, t)$$

This kind of problem brings with itself some problems, mainly that you cannot use the theory of dynamical systems. Let's see why.

⁴Not immediately obvious:

$$\frac{\partial \phi^{t+s}(x_0)}{\partial s} = \frac{\partial \phi^{t+s}(x_0)}{\partial(t+s)} \frac{\partial(t+s)}{\partial s} = \frac{\partial \phi^{t+s}(x_0)}{\partial(t+s)}$$

Now, if we evaluate that at $s = 0$:

$$\frac{\partial \phi^{t+s}(x_0)}{\partial s} \Big|_{s=0} = \frac{\partial \phi^{t+s}(x_0)}{\partial(t+s)} \Big|_{s=0} = \frac{\partial \phi^t(x_0)}{\partial t}$$

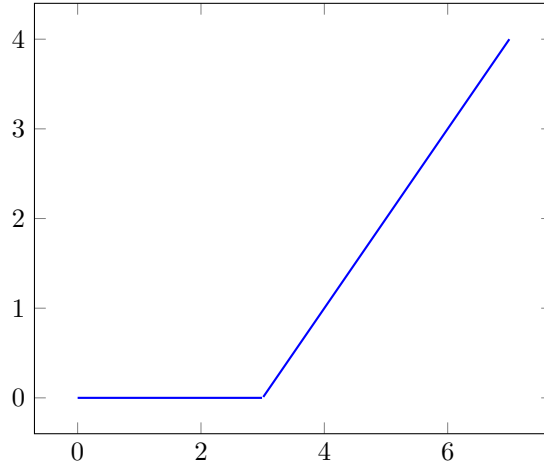


Figure 1.5: Plot of $f(x, t)$ for the example.

Example 1.4 Let's say that:

$$f(x, t) = \begin{cases} 0 & 0 < x < 3 \\ x - 3 & 3 < x \end{cases}$$

Calling the flow $\phi^t(x_0) = x(t, x_0)$, we could easily see that the semi-group property is not satisfied. Say that you apply ϕ^2 twice:

$$\phi^2(x_0) = x_0 \Leftrightarrow \phi^2(\phi^2(x_0)) = \phi^2(x_0) = x_0$$

quite trivially:

$$\phi^4(x_0) = x_0 + 1 \Leftrightarrow \phi^4 \neq \phi^2(\phi^2(x_0))$$

This renders absolutely impossible to use DS theory, at first glance. How do you go around it?

The first way one could think of is to notice that the IVP's initial condition is really not that much of a constraint. Let's see why this is not true.

Proposition 1.4 *We define an IVP as follows:*

$$\begin{cases} \dot{x} = f(x) \\ x(t_0 \neq 0) = x_0 \end{cases}$$

we define $\phi^{t_0, t}(x_0) = x(t, x_0, t_0)$. In this case, you lose the semi-group property, i.e. $\phi^{t, s} \circ \phi^{t_0, t} = \phi^{t_0, s}$.

Proof. Homework. □

There is a correct workaround, and it is the following.

Proposition 1.5 Suppose you have the following IVP, where $x \in M \subseteq \mathbb{R}^d$:

$$\begin{cases} \dot{x} = f(x, t) \\ x(0) = x_0 \end{cases}$$

Which is built on a non-autonomous ODE. We set $y = (x, t) \in M \times \mathbb{R} \subseteq \mathbb{R}^{d+1}$ and $F(y) = F(x, t) = (f(x, t), 1)$.

Then, we consider the IVP:

$$\begin{cases} \dot{y} = F(y) \\ y(0) = (x_0, 0) \end{cases}$$

Where the dot is the derivative with respect to the newly defined "time" s , for which the following is true:

$$\begin{aligned} \begin{bmatrix} \frac{dx}{ds}(s) \\ \frac{dt}{ds}(s) \end{bmatrix} &= \begin{bmatrix} f(x(s), t(s)) \\ 1 \end{bmatrix} \\ \begin{bmatrix} x(0) \\ t(0) \end{bmatrix} &= \begin{bmatrix} x_0 \\ 0 \end{bmatrix} \end{aligned}$$

Then this is an autonomous ODE, with respect to t .

Proof. First things first, we can actually see how s behaves in respect to t from the bottom members of the matrices of its definition:

$$\frac{dt}{ds}(s) = 1, t(0) = 0 \Leftrightarrow t(s) = s$$

so s and t are actually the same. So, by looking at the top members, you find the previous problem with respect to t . $\square$

Observation 1.1 What we are doing is the classic physics trick to consider time as a dimension as the other ones.

So, basically, if you have a non-autonomous ODE you can make it an autonomous one.

1.3 Fixed and periodic points

Definition 1.4 Considering the usual $\mathbb{G} = \mathbb{R}, \mathbb{R}_0^+, \mathbb{Z}, \mathbb{N}$, for a DS $\phi^t_{t \in \mathbb{G}}$, the orbit or trajectory of $x \in M$ is defined as:

$$O(x) = \{\phi^t(x) | t \in \mathbb{G}\}$$

In the cases of interest, we will use

$$O^{t(-)}(x) = \{\phi^t(x) | t \geq (\leq) 0\}$$

and call it future or forward (past or backward, in the case with minus and $\leq$) orbit or trajectory.

Definition 1.5 $\bar{x} \in M$ is called a fixed point if $\phi^t(\bar{x}) = \bar{x}, \forall t > 0$.

Else, $\bar{x}$ is called a periodic point if $\exists T > 0 | \phi^T(\bar{x}) = \bar{x}$, and T is called a period for $\bar{x}$.

In the case of maps, the set of fixed points is sometimes denoted $Fix(T) = \{x \in M | T(x) = x\}$.

Observation 1.2 Clearly, in periodic cases $\phi^{t+T}(\bar{x}) = \phi^t(\bar{x})$.⁵

Observation 1.3 Notice that if $x \in Fix(T)$, then $T^n(x) = x$ which is true if and only if $x \in Fix(T^n)$; more formally:

$$x \in Fix(T) \Rightarrow T^n(x) = x \Leftrightarrow x \in Fix(T^n)$$

Also, one can very easily notice that a periodic point of T is by definition a point in $Fix(T^n)$ for some $n \in \mathbb{Z}$.

1.3.1 Graphical method for the orbit of a 1D map and finding fixed points

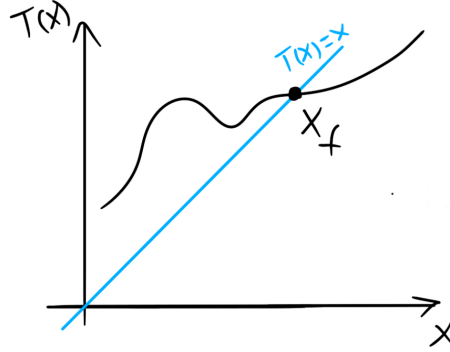


Figure 1.6: Plot of an arbitrary invertible map and its intersection with the bisectrix of the first and third plane, where x_f is a fixed point.

Example 1.5 (1D maps) Consider the following kind of map:

$$T : M \subseteq \mathbb{R} \rightarrow M$$

We want to find the fixed points. By definition, what you are trying to find is the points for which $x = T(x)$. Graphically, this means that by plotting $T(x)$ and then the bisectrix of the first and the third plane, and then finding their intersections, you obtain the fixed points, as shown in Figure 1.6.

Riffing from this simple method, one can develop a really easy graphical method to find the orbit of a 1D map, which works as follows, as shown in Figure 1.7:

⁵Provable by group property on the definition.

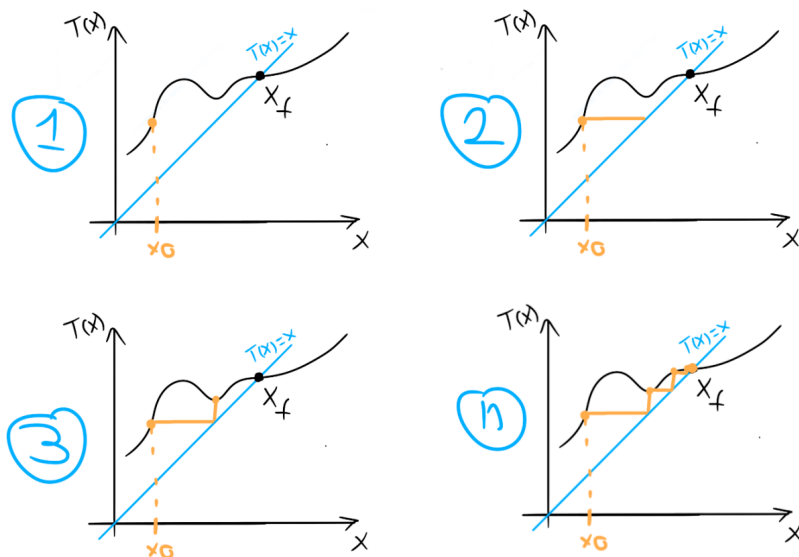


Figure 1.7: Representation of the algorithm after the first steps and when fully executed, on an arbitrary map.

1. Find $T(x_0)$ on the graph;
2. Draw a horizontal line from $T(x_0)$ to the bisectrix of the first and third plane;
3. From the intersection of the bisectrix and the just drawn horizontal line, draw a vertical one until intersecting $T(x)$;
4. Repeat.

Observation 1.4 In the represented case, we find that the orbit converges to the fixed point, but this is not always true, as shown in Figure 1.8. Just from the two examples we did, we may already infer that how the orbit behaves may have something to do with the derivative of the map. We will look further into that during this course.

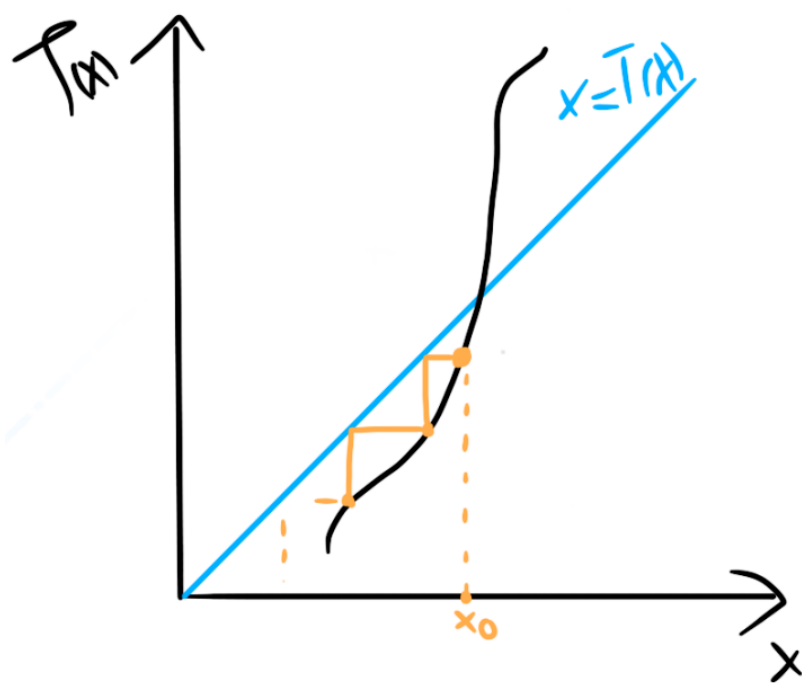


Figure 1.8: Representation of the algorithm after some steps executed on another map.

Chapter 2

Topological dynamics

Chapter 3

Ergodic theory

Chapter 4

Possible developments